25 June 2024

Ref: CL/4475

Subject: **First consultation on the implementation of the 2021 Recommendation on Open Science**

Madam/Sir,

Under Article VIII of UNESCO's Constitution, Member States are required to submit regular reports on the measures they have taken to implement the conventions and recommendations adopted by the Organization.

In this context, I invite your Government to submit its report on the 2021 Recommendation on Open Science, in English or French, **no later than 28 February 2025**. Attached you will find guidelines to assist your authorities in submitting the report. Regional online sessions will also be organized to provide further guidance on report preparation.

Member States are encouraged to organize all necessary consultations with relevant ministries and institutions as well as with key stakeholders, including professional associations, civil society partners, private sector entities and National Commissions for UNESCO.

In addition, Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO regarding reports on the 2021 Recommendation on Open Science. Member States should inform the Secretariat of this designation by 20 July 2024 at the latest at: openscience@unesco.org.

The Secretariat will present the first global report on Member States' application of this Recommendation for examination at the 222nd session of the Executive Board in autumn 2025. This report, together with the comments of the Committee on Conventions and Recommendations, will subsequently be submitted to the 43rd session of the General Conference in 2025.

The Natural Sciences Sector remains at your disposal for any further information you may require.

Please accept, Madam/Sir, the assurances of my highest consideration.

Audrey Azoulay
Director-General

Encl.: *Guidelines for the preparation of the reports from Member States on the implementation of the 2021 Recommendation on Open Science*

cc: Permanent Delegations to UNESCO
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To Ministers responsible for relations with UNESCO

ANNEX

GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as "the 2021 Recommendation") that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i.e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO's Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at https://www.unesco.org/en/open-science/toolkit and https://unesdoc.unesco.org/ark:/48223/pf0000387324.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

(i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted online.

(ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country: Australia
- Organization(s) or entity(ies) responsible for the preparation of the report:

  Department of Industry and Science and Resources

  Website: www.industry.gov.au

  Email address:

  Phone number:

  Please describe the role/mandate of your organization: The Australian Government Department of Industry, Science and Resources delivers the Government's economic agenda by enabling a productive, resilient and sustainable economy, enriched by science and technology.

- Officially designated contact person(s) who completed this survey:

Full name: Stephanie Julienne

Position: Manager, International Science and Resources Branch, International Trade and National Security Division.

Email address:

- Full Name(s) of designated official(s) certifying the report: Stephanie Julienne, Manager, International Science and Resources Branch, International Trade and National Security Division

- Brief description of the consultation process established for the preparation of the report: Consultation with a range of relevant Australian organisations listed in the table below.

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Australian Research Council (ARC)	The National Health Medical Research Council (NHMRC)	Office of the Chief Scientist (OCS)
info@arc.gov.au	reception@nhmrc.gov.au	Contact on their website.
Public Sector	Public Sector	Public Sector
Department of Education	Academy of Social Sciences in Australia (ASSA)	Department of Health and Aged Care (DHAC)
Contact on their website.	info@socialsciences.org.au	Contact on their website.
Public Sector	Non-for-Profit Organisation	Public Sector

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

Yes. The Australian Open Science Network has adopted the UNESCO definition of open science and is working to promote the Recommendation.

The Australian Research Council's (ARC) Open Access Policy [PDF] is currently being reviewed to ensure it is appropriate and up to date. The values and principles of open science are being considered as a part of the policy review process.

The UNESCO recommendation was a key point of the Australian Roadmap for Open Research roundtable of 13 June 2024, held by the Academy of Social Sciences in Australia.

The Medical Research Future Fund (MRFF) open science policy is embedded in all published MRFF grant opportunity guidelines. The grant opportunities are available at grantsconnect.gov.au.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES. English.

1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

YES. Organisations like Open Access Australasia (OAA) and the Council of Australian University Librarians (CAUL) actively promote open access. They advocate for policies that align with international standards like those from UNESCO. Australia's former Chief Scientist, Dr Cathy Foley, actively advocated for open access to research literature during her tenure. She released a report that makes the case for a public access model that would provide access to academic journal articles (including international-authored articles and the back catalogues) for all Australians, as well as open access publishing of all Australian led research so that it is free for readers to access globally.

The Australian Open Science Network has held several webinars that raise awareness on open science.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[ii], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

YES. National Health and Medical Research Council's (NHMRC) Open Access Policy and Research Quality Strategy are aligned with the values and principles of open science. The Open Access Policy mandates that publications are made immediately open access with a Creative Commons Attribution "CC BY" licence and encourages data to be made available with open licencing. The Open Access Policy encourages bibliodiversity through supporting all routes to open access, with no preference for one. The Open Access Policy specifically addresses research involving Aboriginal and Torres Strait Islander people and communities, providing guidance on meeting open access requirements for publications while considering the values of justice, equity, respect and responsibility and the principles of Aboriginal and Torres Strait Islander self-determination and leadership, sustainability and accountability.

NHMRC underwent a public consultation process in April-May 2021 about the proposed changes to its Open Access Policy to require immediate open access and CC BY licencing.

MRFF grant recipients are required to ensure that research publications (and associated research data) are openly accessible, except where publication may compromise the organisation's obligations with respect to patient privacy, intellectual property and/or commercially confidential information.

Additionally, the National Open Science Taskforce (NOST) was formed to represent key research stakeholders and advocate for sectoral ownership and leadership on matters of Open Science. Please view attached document for more information.

PDF
Consultation on
2021 UNESCO Recor

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy[iii], strategy or plan of action on science, technology and innovation (STI)?

YES. Australia does have several national policy strategies and action plans concerning science, technology, and innovation. These policies include the open science elements.

1. Australia's National Science and Research Priorities [PDF]

2. National Science Statement [PDF]
3. AI Action Plan [PDF]
4. National Quantum Strategy [PDF]
5. Australia in Space: A Decadal Plan for Australian Space Science 2021–2030 [PDF]
6. Defence Innovation, Science and Technology Strategy [PDF]
7. 2021 National Research Infrastructure Roadmap [PDF]
8. Australia 2030: Prosperity through Innovation [PDF]

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[iv] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

UNDER DEVELOPMENT. While there isn't a single policy at the national level that addresses open science, the Australian Government's 2024 National Science Statement calls for enhanced efforts to "Modernise our science agency systems and decision-making mechanisms, including to better support open science and cross disciplinary and cross institution collaborations".

NHMRC's Open Access Policy [PDF] uses the UNESCO definition of open science and is aligned with the values and principles of open science. Publications are required to be immediately open access with a CC BY licence and data is encouraged to be made available with open licencing.

The ARC has an Open Access Policy [PDF] which applies to all Research Outputs that come from ARC Funded Research, and their Metadata. Any Research Outputs arising from an ARC research Project must be made accessible within 12 months from the date of publication.

Title	NHMRC Open Access Policy	ARC Open Access Policy
Authority	National Health Medical Research Council (NHMRC)	Australian Research Council (ARC)
Entities involved	The policy was developed and updated through consultation with the Australian medical research sector, including advice from NHMRC's committees.	The policy was developed and updated through consultation with the Australian research sector, including advice from ARC's committees.
Year of adoption	2015	2013
Last updated	2023	2021
Key areas/ sections of policy	Publications and research data: policy and guidance for implementation.	Publications and research data: policy and guidance for implementation.
Elements of Open Science addressed	Publications and research data.	Publications and research data.

2.3 In your country, are there specific policy instruments[v] that aim to promote open science?

UNDER DEVELOPMENT. While there isn't a single policy instrument titled "Open Science Policy," these various initiatives collectively serve to promote and advance open science practices in Australia. The landscape is characterised by a combination of mandates from funding bodies, institutional policies, national strategies, and advocacy efforts, all contributing to a culture of open science.

- Open Access Mandates from Funding Bodies, for example, Medical Research Future Fund (MRFF), National Health and Medical Research Council (NHMRC) and Australian Research Council (ARC).
- National Strategies and Statements including the Australian Government's 2024 National Science Statement, Australia 2030: Prosperity through Innovation and National Health and Medical Research Strategy.
- Open Data Initiatives: Data.gov.au
- Institutional Policies: Australian universities have their own open science or open access policies, which mandate or encourage researchers to make their work openly available.
- Government Initiatives and Proposals: Chief Scientist's Proposal - Dr. Cathy Foley (former Chief Scientist) has proposed a public access model that would involve national read and publish agreements with publishers. Dr Foley’s advice will be considered through appropriate mechanisms.

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

UNDER DEVELOPMENT. The Australian Government is undertaking a strategic examination of our research and development system, to be completed by December 2025. This review, along with the recent advice on open access models (by former Australian Chief Scientist, Dr Cathy Foley) may inform Australia’s strategic approach to open access.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

YES. In Australia, institutional policies and strategies for promoting open science are multifaceted, involving various stakeholders across academia, government, and research funding bodies. Examples include:

- Open Access Australasia
- Council of Australian University Librarians (CAUL)
- Australian Library and Information Association (ALIA)
- Universities around Australia have a range of differing policies on Open Access, but they all support UNESCO’s Recommendation.
- Funding bodies including Australian Research Council (ARC) and National Health Medical Research Council (NHMRC).

NHMRC’s Open Access Policy encourages research institutions to have or develop an institutional open science policy consistent with its Open Access Policy that addresses open access to both publications and data. NHMRC also encourages research institutions to consider rewarding open science practices as part of their recruitment, evaluation and promotion processes.

The MRFF embeds open science policy in its grant opportunity guidelines, including requiring that grant recipients ensure that research publications are openly accessible and that de-identified research data following completion of the grant is in an open access repository and in accordance with best practice.

2.6 In your country, are there specific funding mechanisms[vi] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

YES. There are several funding mechanisms in Australia that support aspects of open science, even if not all are explicitly labelled as "open science" funding. Examples include:

- National Health and Medical Research Council (NHMRC)
- Australian Research Data Commons (ARDC)
- National Collaborative Research Infrastructure Strategy (NCRIS)
- Australian Research Council (ARC)
- Medical Research Future Fund (MRFF)

NHMRC grant funds may be used to support costs associated with publications and open access.

2.7 In your country, have open science practices[vii] been integrated in existing research funding mechanisms?

YES. In Australia, open science practices have been partially integrated into existing research funding mechanisms, but the extent and consistency vary. Examples include:

- Australian Research Council (ARC)
- National Health and Medical Research Council (NHMRC)
- Medical Research Future Fund (MRFF)

The ARC's current contribution is through their Open Access Policy which encourages open science practices.

Open Science practises are mandated (publications) or strongly encouraged (data) for all recipients of NHMRC funding. NHMRC grant funds may be used to support costs associated with publications and open access.

MRFF grantees are required to adhere to open science practices wherever possible, except for where this would present issues around privacy and confidentiality. Grantees are also strongly encouraged to publish de-identified research data and associated metadata in an open access repository or a public database and in accordance with best practice.

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? *(ref.: (ii) 17.i)*

YES. Australia has made significant efforts to foster equitable public-private partnerships in open science, primarily through existing infrastructure programs (NCRIS, ARDC) and CSIRO collaborations. By the end of 2025, increased efforts are foreseen, driven by UNESCO reporting, national policy rollout, and regional leadership ambitions, likely resulting in pilot projects and draft policies.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy *(ref.: (ii) 17.j, 23.c)*

UNDER DEVELOPMENT. Australia has policies, advocacy, and collaborative networks that support open science, and it is envisaged that over time they could lead to development of a national monitoring framework for open science.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. *(ref.: (iii) 18.a)*

The 2025 Strategic Examination of R&D discussion paper reported, 1.66% of Australian GDP was dedicated to Research Development Expenditure in 2022.

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. *(ref.: (iii) 18.b)*

In 2024, with a current population of 26.6 million, 94.9% of adult Australians are internet users with reliable internet and bandwidth.

3.3 Are there national research and education networks (NRENs)[viii] active in the country? *(ref.: (iii) 18.c)*

YES. In Australia, the national and international NREN operator is the Australian Academic and Research Network (AARNet).

3.4 Are there national or institutional open science infrastructures[ix] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

YES. There are a range of national and institutional open science infrastructures in Australia. Australia's National Collaborative Research Infrastructure Strategy program funds and coordinates open access research infrastructure facilities across a range of specialities. These capabilities are outlined at riconnected.org.au. The Australian Government also funds and operates landmark infrastructure such as the Australian Synchrotron and hosts global infrastructure such as the Square Kilometre Array project. Some useful examples of NCRIS-funded activity is the development of Research Link Australia, is a searchable database of researchers, research projects and organisations across Australia that can be filtered by research area.

In Australia, there are several Australian institutions such as the National Library of Australia and universities independently hosting several open access journals. Although there are no national platforms, the open access journals hosted by these Australian Institutes are community managed infrastructures which meets the needs of local disciplines.

Australian national and some shttps://www.ipc.nsw.gov.au/information-access/open-government-open-data have adopted Open Data policies (for non-sensitive data) and have also committed to open licenses as a default for their own policy and information outputs.

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? *(ref.: (iii) 18.e, 18.f)*

YES. While Australia does not host a single infrastructure, named "federated regional or international IT infrastructure" for open science, its participation in, and contributions to, various networks, projects, and initiatives effectively create a federated environment where data and resources are shared both nationally and globally for the advancement of open science. Many of these facilities are supported by the NCRIS program.

National Contributions to International Infrastructures

Open Science Initiatives
The Global Sustainability Coalition for Open Science Services (SCOSS)
While not directly hosted by Australia, Australian institutions like the Australian National University and Queensland University of Technology have been involved in supporting SCOSS through funding and advocacy. SCOSS helps secure funding for critical open science infrastructure globally since 2017.

Federated Regional Infrastructures
Australian Academic and Research Network (AARNet)
Although primarily national, AARNet serves as Australia's gateway to international networks, effectively creating a federated environment where Australian researchers can seamlessly interact with global research infrastructures. Established in 1989, AARNet's connections extend to various international research and education networks, enhancing federated access to IT resources for open science.
Australian Research Data Commons (ARDC)
ARDC operates Research Data Australia (RDA), which not only aggregates metadata from Australian data providers but also aligns with international standards for data sharing. This platform supports the principles of federated access by making Australian research data discoverable and accessible to both national and international researchers.

International Research Collaboration
Square Kilometre Array (SKA)
While the SKA itself is an international project, Australia hosts part of the infrastructure (the ASKAP telescope) since 2012 and is involved in the development of SKA's data processing capabilities. This project exemplifies federated international collaboration in science and technology, with data being shared across continents.

Funding and Hosting:
Australian Access Federation (AAF)
AAF facilitates federated identity management, allowing researchers from different institutions to access shared resources securely. This supports the infrastructure for open science by simplifying access to data and services across institutions and, by extension, internationally.
National Collaborative Research Infrastructure Strategy (NCRIS)
NCRIS provides funding for various projects that could be considered part of a federated infrastructure for open science. For instance:

- Bioplatforms Australia supports the sharing of biological data in an open science framework, contributing to international genomic data resources.
- The Atlas of Living Australia (ALA) participates in global biodiversity data initiatives, contributing Australian data to the Global Biodiversity Information Facility (GBIF).
- Terrestrial Ecosystem Research Network (TERN)
- Integrated Marine Observing System (IMOS)

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? *(ref.: (iii) 18.g)*

YES. Australia is deeply engaged in North-South, North-South-South, and South-South collaborations to optimise infrastructure use and advance open science platforms. Through partnerships like Landsat Next, Southern Ocean Observing System (SOOS), and regional initiatives (including CSIRO's Regional Partnerships) in the Pacific and Southeast Asia. Australia leverages its position to bridge northern and southern interests. These efforts not only enhance scientific capacity and infrastructure efficiency but also promote equitable access to knowledge and resources, aligning with global goals for sustainable development and regional prosperity.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? *(ref.: (iv) 19.a, 19.c)*

YES. The 2021 National Research Infrastructure Roadmap calls out the importance of workforce development for open research infrastructure. This roadmap will continue to inform investments throughout 2025, before a new roadmap is developed for 2026.

4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

YES. The 2021 National Research Infrastructure Roadmap calls out the importance of workforce for open research infrastructure. This roadmap will continue to inform investments throughout 2025, before a new roadmap is developed. National Collaborative Research Infrastructure Strategy (NCRIS) also supports research infrastructures integrates core values and principles of Open Science.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? *(ref.: (iv) 19.b)*

UNDER DEVELOPMENT. The Universities Accord Panel's Final Report recommends open data (which supports open science) be leveraged to support research evaluation processes (p 220), although the Australian Government has not yet released a response to this recommendation.

NHMRC is developing guidance on the education and training of researchers about good research practices, including open science.

4.4 Are open educational resources [x] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? *(ref.: (iv) 19.d)*

YES.

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? *(ref.: (iv) 19.e)*

YES. The Australian Government's 2024 National Science Statement calls for enhanced efforts to **"**Modernise our science agency systems and decision-making mechanisms, including to better support open science and cross disciplinary and cross institution collaborations".

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

YES. The Universities Accord Panel's Final Report recommends open data (which supports open science) be leveraged to support research evaluation processes (p 220).

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? *(ref.: (v) 20.a, 20.c, 20.d)*

NO. In Australia, the incorporation of specific clauses from the UNESCO Recommendation on Open Science particularly those recognising open science practices such as sharing, collaborating, engaging with societal actors beyond the scientific community, and fostering dialogue with other knowledge systems, into career evaluation and progression systems is underway but remains incomplete. Many institutions, including UNSW, now adopt societal impact frameworks.

Progress reflects Australia's commitment to open science, as evidenced by national policies and institutional reforms, yet full integration into career frameworks is still evolving.

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? *(ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)*

YES. There are several initiatives in Australia aimed at rewarding and promoting open science practices, including:

Open Science Badges: Some Australian journals have adopted the use of Open Science Badges, which are visual indicators that acknowledge researchers for sharing data, materials, or for pre-registering their studies. These badges are used by journals to signal and reward open practices, although this is more on a journal-to-journal basis rather than a widespread national initiative.

Incentives through Funding Bodies: NHMRC has policies that support open access and open data, with a mandate for immediate open access for NHMRC-funded research. While not explicitly an award, this policy incentivises open practices by making it a condition of funding. Similarly, ARC requires funded research to be made open access within 12 months, which indirectly rewards open science by setting expectations and mandates.

NHMRC is a signatory of the 2012 San Francisco Declaration on Research Assessment and the Agreement on Reforming Research Assessment and are members of the Coalition for Advancing Research Assessment (CoARA). NHMRC is developing an action plan aligned with the CoARA core commitments. NHMRC funding schemes that assess publication history require applicants to not include publication metrics such as journal impact factors and provide guidance to peer reviewers on assessing the quality of research publications.

NHMRC's Open Access Policy encourages research institutions to consider rewarding open science practices as part of their recruitment, evaluation and promotion processes. NHMRC is developing a Good Institutional Practice Guide, to provide guidance to NHMRC-funded institutions and NHMRC-funded researchers about good institutional practice to promote open, honest, supportive and respectful institutional cultures conducive to the conduct of high-quality research. The Guide encourages recognising and rewarding open science practices.

5.4 Have there been any unintended negative consequences[xi] of open science practices reported in your country? (ref.: (v) 20)

NO.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

YES, at both national and institutional levels.

National Level:

Former Chief Scientist, Dr Foley published a report and hosted a webinar outlining her advice on open access models, which has been submitted for consideration by the Australian Government.

NHMRC's Open Access Policy encourages the posting of preprints on preprint servers with open licensing. NHMRC accepts preprints in grant applications. NHMRC's Open Access Policy strongly encourages grant recipients to take reasonable steps to share research data and associated metadata arising from NHMRC-funded research.

NHMRC's Targeted Calls for Research funding scheme enables community and professional groups to submit research topics that may be underfunded or have a significant research knowledge gap. Consumer and community representatives are routinely appointed to NHMRC Council and Principal Committees and, where appropriate, working committees. NHMRC is trialing the involvement of consumer and community reviewers in peer review for select grant opportunities.

For some MRFF grant opportunities the policy underlying dissemination of research outcomes linked to clinical trials include (with the exception where publication may compromise obligations with respect to patient privacy, intellectual property and/or commercially confidential information) grantees are required to:

- if a clinical trial, submit the clinical trial protocol to an open access repository within six months of Human Research Ethics Committee (HREC) approval, or publish a protocol manuscript as soon as practicable.
    - within 12 months of completion of the grant activity, disseminate the research findings through:
- ensuring that research findings are available in an open access repository.
    - content specific forums
    - submission to peer-reviewed journals.
- make lay summaries available to research participants, concurrently with sharing and dissemination of research results.

Institutional Level:

The CAUL Open Education Resources Collective enables number of Open Educational Resources (OERs), published through PressBooks.

Australian Universities have their own policies which encourages exploration of open peer-review practices, sharing of negative results and associated data, and promote open innovative practices to increase the openness of the scientific process.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? *(ref.: (vii) 22.a)*

The Universities Accord Panel's Final Report highlights the importance of coordinating open access (to national research infrastructure) to enable Australia to adapt to technological shifts.

Australia's Nuclear Science and Technology Organisation (ANSTO) engages multilaterally through the International Atomic Energy Agency (IAEA) and regionally through the Regional Cooperative Agreement for Research, Development and Training Related to Nuclear Science and Technology for Asia and the Pacific (RCA) and the Forum for Nuclear Cooperation in Asia (FNCA). The RCA is a Treaty level agreement between 22 countries in Asia and the Pacific under the auspices of the International Atomic Energy Agency (IAEA). ANSTO also engages bilaterally with similar counterparts around the world.

Australia has established partnerships across various universities (Australian and international) collaborating across joint scientific research including participation in the Council of Australasian University Librarians (CAUL), which includes most of Australia's universities and major Australian Government-funded research agencies such as the CSIRO, Defence Science and Technology Group (DSTG) and ANSTO, and has negotiated open access agreements with over 20 publishers since 2020.

The ANSTO Open Science Portal website provides access to research data, and ANSTO also hosts the Australian IAEA International Nuclear Information System (INIS) with a statutory responsibility to identify, capture and index relevant Australian literature in the field of nuclear science and technology for INIS for free global access and engages in multidisciplinary and international research (p 206).

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? *(ref.: (vii) 22.b, 22.e, 22.f)*

Under Development. Relevant Australian Government agencies plan and collaborate on open science. There is currently no singular national strategy on this.

NHMRC are members of several international working groups, committees and communities of best practice related to open science and research assessment. These include the Asia-Pacific DoRA Funders working group, cOAlition S Leaders and Experts committees, and CoARA working groups. NHMRC has engaged with UNESCO Open Science working groups.

MRFF has representatives that engage with DoRA Funders working groups and attend regular meetings within this context.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=21)

YES. Australia is a member of several international organisations and working groups that provide funding mechanisms for Open Science. These include DoRA Funders and CoARA working groups. Regional organisations included MRFF, NHMRC, ACR.

In addition, Open Access Australasia engages with and promotes to the university sector the activities and approaches of international organisations such as Invest in Open Infrastructure and the SCOSS. CAUL also actively promotes SCOSS and their initiatives.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

NO.

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i Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

xi For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.